Python Code


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    "# There are two types of Data Structures\n",
    "\n",
    "**1. Mutable**\n",
    "\n",
    "Mutable is when something is changeable or have the ability to change. in python, 'mutable' is the ability to change values. These are often objects that store a collection of data. For example:\n",
    "- List[]\n",
    "- sets{}\n",
    "- Dictionaries {key:values}\n",
    "\n",
    "**2. Immutable**\n",
    "\n",
    "It is not possible to alter the value of an object in python, it is known as immutable object. Once Immutable object is created it's value remains permanent and unchangeable.\n",
    "\n",
    "- numerical(int,float)\n",
    "- strings\n",
    "- tuples "
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    "# int float str ---> cannot modify "
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    "print(a)\n",
    "print(id(a))"
   ]
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```

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     "print(id(a))"
    ]
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       "2314249563712\n"
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     "# built in function \n",
     "# id \n",
     "\n",
     "# mutable data structures \n",
     "# list \n",
     "\n",
     "l1 = ['tapsya','krish','yash','yashpal','tanish']\n",
     "\n",
     "print(l1)\n",
     "print(id(l1))"
    ]
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       "2314249563712\n"
      ]
     }
```

```
    ],
    "source": [
     "# add jeet kumar \n",
     "\n",
     "l1.append('jeet kumar')\n",
     "print(l1)\n",
     "print(id(l1))"
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     "\n",
     "- sequence of characters\n",
     "- 'sent',\"sent\", '''paragraphs'''\n",
     "- immutable\n",
     "- ordered\n",
     "- duplicacy allowed"
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       "text/plain": [
        "'tanish'"
       ]
      },
      "execution_count": 9,
      "metadata": {},
      "output_type": "execute_result"
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     "str1 = 'tanish'\n",
     "str1"
    ]
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    "metadata": {},
    "outputs": [],
    "source": [
     "par1 = '''At a noisy bus stop, two strangers shared a smile over spilled tea.\n",
     "Days turned into routines, routines into trust.\n",
     "When life hit hard, they stood side by side, saying little.\n",
     "Friendship, they learned, isn't about constant laughter—it's about showing up, even in silence.\n",
     "'''"
    ]
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       "Days turned into routines, routines into trust.\n",
       "When life hit hard, they stood side by side, saying little.\n",
       "Friendship, they learned, isn’t about constant laughter—it's about showing up, even in silence.\n",
       "\n"
      ]
     }
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     "par1\n",
     "print(par1)"
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     "\n",
     "# indexing \n",
     "# slicing \n",
     "# moreover, same order as passed "
    ]
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        "'r'"
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     "str2 = 'krish'\n",
     "str2\n",
     "str2[1]"
    ]
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```
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       "text/plain": [
        "'111111111111'"
       ]
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     "\n",
     "str2 ='111111111111'\n",
     "str2"
    ]
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     "\n",
     "# upper \n",
     "# lower\n",
     "# swapcase "
    ]
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        "'JEET'"
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     "str1.upper()"
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```

```
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    "str1"
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     "data": {
      "text/plain": [
       "'jeet'"
      ]
     },
     "execution_count": 26,
     "metadata": {},
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   "source": [
    "str1.lower()"
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   "metadata": {},
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       "'I am Krish. I like to study math. I am straight from an Action Movie'"
      ]
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     "execution_count": 28,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "sent1 = 'i AM kRISH. i LIKE TO STUDY MATH. i AM STRAIGHT FROM AN aCTION mOVIE'\n",
    "sent1.swapcase()"
   ]
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```

```
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     "# split \n",
     "# strip \n",
     "# lstrip \n",
     "# rstrip \n",
     "# replace"
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       ]
      },
      "execution_count": 31,
      "metadata": {},
      "output_type": "execute_result"
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    ],
    "source": [
     "# strip \n",
     "\n",
     "a = 'jeet'\n",
     "b = ' jeet  '\n",
     "\n",
     "a==b"
    ]
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    "outputs": [
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      "data": {
       "text/plain": [
        "' jeet'"
       ]
      },
      "execution_count": 35,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
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     "# lstrip \n",
     "b.lstrip()\n",
     "b.rstrip()"
    ]
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```

```
    "outputs": [
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       "text/plain": [
        "'anshum'"
       ]
      },
      "execution_count": 38,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# special characters remove \n",
     "\n",
     "c = '!!?anshum!!!'\n",
     "c.strip('!?')"
    ]
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    "metadata": {},
    "outputs": [],
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     "\n",
     "str4 = 'jeet iqra yash mohit yashpal saksham'"
    ]
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      "data": {
       "text/plain": [
        "'yash'"
       ]
      },
      "execution_count": 61,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "str4\n",
     "str4[:4]\n",
     "str4[5:9]\n",
     "\n",
     "str4.split()[2]"
    ]
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    "metadata": {},
    "outputs": [
     {
```

```
      "data": {
       "text/plain": [
        "['data analytics', 'data science', 'digital marketing']"
       ]
      },
      "execution_count": 50,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# split()\n",
     "# delimeter : -,|\n",
     "\n",
     "cs = 'data analytics/data science/digital marketing'\n",
     "cs.split('/')"
    ]
   },
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    "metadata": {},
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      "data": {
       "text/plain": [
        "'iqra likes to study math'"
       ]
      },
      "execution_count": 53,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# replace \n",
     "\n",
     "sent1 = 'krish likes to study math'\n",
     "sent1.replace('krish','iqra')"
    ]
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      "data": {
       "text/plain": [
        "'anshum'"
       ]
      },
      "execution_count": 54,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "sent1 = '!!anshum!'\n",
     "sent1.replace('!','')"
    ]
```

```
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   "outputs": [],
   "source": [
    "# index \n",
    "# count \n",
    "\n",
    "lname = '''Prabhakarna Sripalawardhana Atapattu Jayasuriya Laxmansriramkrishna Shivavenkata Rajasekara Sriniwasana Trichipalli Yekya Parampeel Parambatur Chinnaswami Muthuswami Venugopal Iyer.'''"
   ]
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      ]
     },
     "execution_count": 63,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
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    "lname.index('Rajasekara')"
   ]
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      "text/plain": [
       "'Rajasekara'"
      ]
     },
     "execution_count": 66,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "lname[81:81+10]"
   ]
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```

```
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        "40"
       ]
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      "output_type": "execute_result"
     }
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     "lname.lower().count('a')\n",
     "# lname.count('i')"
    ]
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     "# slicing\n",
     "# variable_name[start:end:step_size]"
    ]
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    ]
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       "text/plain": [
        "'raj'"
       ]
      },
      "execution_count": 76,
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      "output_type": "execute_result"
     }
    ],
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     "str5[3:6]"
    ]
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```

```
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     "# True and False \n",
     "# isupper \n",
     "# islower \n",
     "# isalpha \n",
     "# isnumeric\n",
     "# isalphanumeric \n",
     "# isprintable \n",
     "# isspace"
    ]
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       "text/plain": [
        "False"
       ]
      },
      "execution_count": 83,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
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     "str1.isupper()"
    ]
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        "True"
       ]
      },
      "execution_count": 84,
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      "output_type": "execute_result"
     }
    ],
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     "str1 = 'bnuibf'\n",
     "str1.islower()"
    ]
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```

```
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       "text/plain": [
        "True"
       ]
      },
      "execution_count": 88,
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      "output_type": "execute_result"
     }
    ],
    "source": [
     "# isalpha --> true --> alphabets \n",
     "# no special characters, no spaces, no numbers \n",
     "\n",
     "name = 'anshum'\n",
     "name.isalpha()"
    ]
   },
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        "True"
       ]
      },
      "execution_count": 89,
      "metadata": {},
      "output_type": "execute_result"
     }
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     "\n",
     "str1 = '4332553'\n",
     "str1.isnumeric()"
    ]
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       "text/plain": [
        "True"
       ]
      },
      "execution_count": 92,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# isalnum\n",
```

```
    "\n",
    "str1 = 'anshum7banga'\n",
    "str1.isalnum()"
   ]
  },
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       "False"
      ]
     },
     "execution_count": 95,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
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    "# isprintable \n",
    "str1 = 'sfnbuiorfw\\n'\n",
    "str1.isprintable()"
   ]
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     "text": [
      "   \n"
      ]
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    "x = '   ' # detection is difficult \n",
    "print(x)"
   ]
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       "True"
      ]
     },
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     "metadata": {},
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```

```
      }
     ],
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      "x.isspace()"
     ]
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      "# list \n",
      "\n",
      "- multiple elemenents\n",
      "- enclosed []\n",
      "- mutable\n",
      "- duplicacy allowed\n",
      "- ordered\n",
      "- multiple data types -- heterogenous"
     ]
    },
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     "execution_count": 1,
     "id": "595ad5dd-08f5-4e77-9733-f72cb2bc34aa",
     "metadata": {},
     "outputs": [],
     "source": [
      "l1 = ['chicken','rajma chawal','rajma chawal','eggs','pizza','chicken','matar paneer']"
     ]
    },
    {
     "cell_type": "code",
     "execution_count": 6,
     "id": "9ade5cbf-2713-401e-ae80-2c6197c40d21",
     "metadata": {},
     "outputs": [
      {
       "data": {
        "text/plain": [
         "['chicken',\n",
         " 'rajma chawal',\n",
         " 'rajma chawal',\n",
         " 'eggs',\n",
         " 'pizza',\n",
         " 'chicken',\n",
         " 'matar paneer']"
        ]
       },
       "execution_count": 6,
       "metadata": {},
       "output_type": "execute_result"
      }
     ],
     "source": [
      "l1[0]\n",
      "l1"
     ]
    },
    {
     "cell_type": "code",
     "execution_count": 5,
     "id": "b45f65ff-e136-4438-977b-f670e308ecae",
```

```
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "[1, 2, 3, 'anshum', 4.5]"
       ]
      },
      "execution_count": 5,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "l2 = [1,2,3,'anshum',4.5]\n",
     "l2"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 7,
    "id": "e020b67e-307a-4f05-bdba-82269f50efd9",
    "metadata": {},
    "outputs": [],
    "source": [
     "# methods \n",
     "\n",
     "# append \n",
     "# insert"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 17,
    "id": "034b40c6-4e32-4d31-864a-376d36b4e6d2",
    "metadata": {},
    "outputs": [],
    "source": [
     "l1 = ['saksham','jeet','krish', 'raghav']"
    ]
   },
   {
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    "execution_count": 18,
    "id": "e9d09cee-8eea-42cd-b056-bb109857e8ec",
    "metadata": {},
    "outputs": [],
    "source": [
     "# append \n",
     "l1.append('arif')"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 20,
    "id": "ec3d40ae-6a66-4d5e-bba8-bdec3962e4c4",
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "['saksham', 'jeet', 'krish', 'raghav', 'arif', 2]"
       ]
```

```
      },
      "execution_count": 20,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "l1.append(2)\n",
     "l1"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 21,
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    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "['saksham', 'Raghav', 'jeet', 'krish', 'raghav', 'arif', 2]"
       ]
      },
      "execution_count": 21,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# insert \n",
     "\n",
     "l1.insert(1,'Raghav')\n",
     "l1"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 22,
    "id": "bf869d38-9476-4b47-af95-93c7f7a8fbc0",
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "['saksham', 'Raghav', 'krish', 'raghav', 'arif', 2]"
       ]
      },
      "execution_count": 22,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# remove \n",
     "# pop \n",
     "# clear\n",
     "\n",
     "l1.remove('jeet')\n",
     "l1"
    ]
   },
   {
    "cell_type": "code",
```

```
    "execution_count": 23,
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    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "'krish'"
       ]
      },
      "execution_count": 23,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# pop \n",
     "l1.pop(2)"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 24,
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    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "['saksham', 'Raghav', 'raghav', 'arif', 2]"
       ]
      },
      "execution_count": 24,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "l1"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 25,
    "id": "c6d800e9-ad23-48b3-b8dc-ea6c19e625e8",
    "metadata": {},
    "outputs": [],
    "source": [
     "# clear \n",
     "\n",
     "l1.clear()"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 26,
    "id": "69366722-ff7d-489f-a7e1-6f16a2240bcc",
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "[]"
```

```
      ]
     },
     "execution_count": 26,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "l1"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 31,
   "id": "a42d5af8-6069-4d1d-85b4-dc37a1da1da3",
   "metadata": {},
   "outputs": [],
   "source": [
    "# changing values\n",
    "l2 = ['raghav','jeet']\n",
    "l2[0] = 'krish'"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 32,
   "id": "2b310de1-2045-4059-99e8-a1b7156da676",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "['krish', 'jeet']"
      ]
     },
     "execution_count": 32,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "l2"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 41,
   "id": "02f995de-ccbf-4ea2-9bad-fbf84371c45e",
   "metadata": {},
   "outputs": [],
   "source": [
    "# index \n",
    "names = [\"Krishna\",\"Rohan\",\"Aryan\",\"Meera\",\"Saanvi\",\"Nisha\",\"Tanya\",\"Kunal\",\"Rahul\"]"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 42,
   "id": "182ddced-660d-4964-8730-06eba72de50d",
   "metadata": {},
   "outputs": [
    {
```

```
      "data": {
       "text/plain": [
        "'Krishna'"
       ]
      },
      "execution_count": 42,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "names.index('Krishna')\n",
     "names[0]"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 43,
    "id": "cd2804db-00a1-4490-a016-eefe22b751e9",
    "metadata": {},
    "outputs": [],
    "source": [
     "# sort \n",
     "# count\n",
     "# copy "
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 50,
    "id": "4b348d9d-519a-45bc-9c6c-b3b4274bbe66",
    "metadata": {},
    "outputs": [
     {
      "name": "stdout",
      "output_type": "stream",
      "text": [
       "[22, 32, 12, 42, 445, 354, 33, 234, 44]\n",
       "\n",
       "after sort\n",
       "ascending\t [12, 22, 32, 33, 42, 44, 234, 354, 445]\n",
       "descending\t [445, 354, 234, 44, 42, 33, 32, 22, 12]\n"
      ]
     }
    ],
    "source": [
     "l2 = [22,32,12,42,445,354,33,234,44]\n",
     "print(l2)\n",
     "print('\\nafter sort')\n",
     "l2.sort()\n",
     "print('ascending\\t',l2)\n",
     "print('descending\\t',l2[::-1])"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 53,
    "id": "9bea0f51-d711-492f-8e8b-02ff31e3fc91",
    "metadata": {},
    "outputs": [],
    "source": [
     "# count \n",
     "\n",
```

```
    "l3 = ['jeet','raghav','jeet','jeet','krish']"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 54,
   "id": "bee40487-ac66-4d6e-afee-029c360ee527",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "3"
      ]
     },
     "execution_count": 54,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "l3.count('jeet')"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 63,
   "id": "78a25679-1b3c-4e7a-86c0-2f2550f77dfc",
   "metadata": {},
   "outputs": [
    {
     "name": "stdout",
     "output_type": "stream",
     "text": [
      "original\t [1, 2, 3, 4, 5, 6, 7, 8, 9]\n",
      "original\t 2499817544576\n",
      "duplicate\t [1, 2, 3, 4, 5, 7, 8, 9]\n",
      "duplicate\t 2499813869696\n"
     ]
    }
   ],
   "source": [
    "# copy \n",
    "\n",
    "l1 = [1,2,3,4,5,6,7,8,9] # original \n",
    "l2 = l1.copy()\n",
    "l2.remove(6)\n",
    "print('original\\t',l1)\n",
    "print('original\\t',id(l1))\n",
    "\n",
    "print('duplicate\\t',l2)\n",
    "print('duplicate\\t',id(l2))"
   ]
  },
  {
   "cell_type": "markdown",
   "id": "6ff9b0a6-1935-4eae-8277-a81a0f3da1a0",
   "metadata": {},
   "source": [
    "# tuples \n",
    "\n",
    "- entrapped ()\n",
    "- immutable\n",
```

```
    "- duplicacy allowed\n",
    "- heterogenous\n",
    "- ordered"
   ]
  },
  {
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   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "tuple"
      ]
     },
     "execution_count": 65,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "t1 = (1,2,3,4,5,6,7)\n",
    "t1\n",
    "type(t1)"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 68,
   "id": "9959d3de-b2aa-4860-94d5-80f3128ca139",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "4"
      ]
     },
     "execution_count": 68,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "t1[3] # index "
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 70,
   "id": "0f68e312-ffd4-4594-96f2-023c032d5750",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "('anshum', 1, 1, 2, 4.4)"
      ]
     },
     "execution_count": 70,
     "metadata": {},
```

```
      "output_type": "execute_result"
     }
    ],
    "source": [
     "t1 = ('anshum',1,1,2,4.4)\n",
     "t1"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 71,
    "id": "fd31d988-030b-48e4-aa82-e9096c4dec7c",
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "0"
       ]
      },
      "execution_count": 71,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# methods \n",
     "# index \n",
     "# count \n",
     "\n",
     "t1.index('anshum')"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 72,
    "id": "4cbdc6e3-5d64-4601-b3b8-ea86e7c54c5b",
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "2"
       ]
      },
      "execution_count": 72,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "t1.count(1)"
    ]
   },
   {
    "cell_type": "markdown",
    "id": "f8862734-c202-4c35-ba35-643faaef3a8a",
    "metadata": {},
    "source": [
     "# dictionaries \n",
     "\n",
     "- {'key':'value'}\n",
     "- duplicacy is not allowed\n",
```

```
    "- mutable\n",
    "- multipe data types allowed\n",
    "- ordered"
   ]
  },
  {
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   "execution_count": 83,
   "id": "01926875-0d76-494d-bec6-97c06904dfb1",
   "metadata": {},
   "outputs": [],
   "source": [
    "dict1 = {\"Pookie\":\"describe someone cute, beloved, or very close\",\n",
    "        'happiness':'the state of being happy.',\n",
    "        'sadness':'the condition or quality of being sad.',\n",
    "        'sadness':'the condition or quality of being sad.'}"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 84,
   "id": "dfd84c95-9f35-454a-b60c-5b22f015b9e5",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{'Pookie': 'describe someone cute, beloved, or very close',\n",
       " 'happiness': 'the state of being happy.',\n",
       " 'sadness': 'the condition or quality of being sad.'}"
      ]
     },
     "execution_count": 84,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "dict1"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 85,
   "id": "f4fa67d0-9d2e-43d9-869a-c6d3f1e028fb",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{'name': 'jeet', 'age': 21, 'marks': 90, 'teeth': 40}"
      ]
     },
     "execution_count": 85,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "data = {'name':'jeet','age':21,'marks':90,'teeth':40}\n",
    "data"
   ]
  },
```

```
  {
   "cell_type": "code",
   "execution_count": 86,
   "id": "7eaf3a29-ed0b-43ed-995e-1ff6dd8a32e5",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{'name': 'jeet', 'age': 21, 'marks': 90, 'teeth': 16}"
      ]
     },
     "execution_count": 86,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "# values \n",
    "# calling \n",
    "\n",
    "data['teeth'] = 16\n",
    "data"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 93,
   "id": "78054919-afba-4744-aec3-6a15791a6714",
   "metadata": {},
   "outputs": [],
   "source": [
    "# update \n",
    "data\n",
    "data.update({'subject':'python','course':'data science'})"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 94,
   "id": "fa2d01e2-c0f1-4822-92af-b263df38a800",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{'name': 'jeet',\n",
       " 'age': 21,\n",
       " 'marks': 90,\n",
       " 'teeth': 16,\n",
       " 'subject': 'python',\n",
       " 'course': 'data science'}"
      ]
     },
     "execution_count": 94,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "data"
   ]
  },
```

```
  {
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   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{'name': 'jeet',\n",
       " 'age': 21,\n",
       " 'marks': 90,\n",
       " 'subject': 'python',\n",
       " 'course': 'data science'}"
      ]
     },
     "execution_count": 96,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "# pop \n",
    "# pop_item \n",
    "data.pop('teeth')\n",
    "data"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 97,
   "id": "1cd91e08-cc86-40e5-aea6-3c8e037b2012",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "('course', 'data science')"
      ]
     },
     "execution_count": 97,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "# popItem\n",
    "data.popitem()"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 98,
   "id": "eeecbdcd-0e88-42ca-a982-781a5ecfd918",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{'name': 'jeet', 'age': 21, 'marks': 90, 'subject': 'python'}"
      ]
     },
     "execution_count": 98,
```

```
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "data"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 99,
    "id": "4df0d9f3-a97c-4319-a833-f09d6ca05a10",
    "metadata": {},
    "outputs": [],
    "source": [
     "# keys \n",
     "# values \n",
     "# items\n",
     "# get \n",
     "# clear"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 100,
    "id": "08e3f403-aff4-4b82-9cbd-67a48d6f781d",
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "dict_keys(['name', 'age', 'marks', 'subject'])"
       ]
      },
      "execution_count": 100,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "data.keys()"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 101,
    "id": "16da426a-bf9f-4ddd-9f8d-dd88256b0b82",
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "dict_values(['jeet', 21, 90, 'python'])"
       ]
      },
      "execution_count": 101,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "data.values()"
    ]
```

```
   },
   {
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    "execution_count": 102,
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    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "dict_items([('name', 'jeet'), ('age', 21), ('marks', 90), ('subject', 'python')])"
       ]
      },
      "execution_count": 102,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "data.items()"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 104,
    "id": "dc85b74f-a9b7-48d1-86c6-f777d03a66c6",
    "metadata": {},
    "outputs": [],
    "source": [
     "# get \n",
     "data.get('name')\n",
     "data.get('fewbwuiefwb')"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 105,
    "id": "8c448707-b393-4581-8760-9eedf0419893",
    "metadata": {},
    "outputs": [],
    "source": [
     "# clear \n",
     "data.clear()"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 106,
    "id": "9c13e052-cb11-46bd-8469-68f4e3d1a9a5",
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "{}"
       ]
      },
      "execution_count": 106,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
```

```
    "data"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 109,
   "id": "7dd65518-0e94-490a-a8ab-f35431d1db6f",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "'describe someone cute, beloved, or very close'"
      ]
     },
     "execution_count": 109,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "# setdefault \n",
    "\n",
    "dict1.setdefault('Pookie')"
   ]
  },
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   "outputs": [],
   "source": [
    "dict1.setdefault('name')"
   ]
  },
  {
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   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{'Pookie': 'describe someone cute, beloved, or very close',\n",
       " 'happiness': 'the state of being happy.',\n",
       " 'sadness': 'the condition or quality of being sad.',\n",
       " 'name': None}"
      ]
     },
     "execution_count": 111,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "dict1"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 112,
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    "id": "1da248f3-b848-4603-a248-ee83aeae79bc",
    "metadata": {},
    "outputs": [],
    "source": [
     "dict1['name']='krish'"
    ]
   },
   {
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    "id": "2dc0c348-5ab2-44cd-be03-b111629d31b1",
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    "outputs": [
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      "data": {
       "text/plain": [
        "{'Pookie': 'describe someone cute, beloved, or very close',\n",
        " 'happiness': 'the state of being happy.',\n",
        " 'sadness': 'the condition or quality of being sad.',\n",
        " 'name': 'krish'}"
       ]
      },
      "execution_count": 113,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "dict1"
    ]
   },
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    "metadata": {},
    "outputs": [],
    "source": []
   },
   {
    "cell_type": "markdown",
    "id": "f02faf6f-10c6-4bac-896c-637e6e455958",
    "metadata": {},
    "source": [
     "# Sets \n",
     "\n",
     "- mutable data type\n",
     "- unordered\n",
     "- duplicacy is not allowed\n",
     "- entrapped {}\n",
     "- multiple data types are accepted"
    ]
   },
   {
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    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "set"
```

```
      ]
     },
     "execution_count": 1,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "s1 = {1,2,3,4,5,6,7}\n",
    "type(s1)"
   ]
  },
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   "id": "efb98ad6-fe5f-4b32-8730-22793b58c30b",
   "metadata": {},
   "outputs": [],
   "source": [
    "s2 = { 'dog' , 'loin' , 'rabbit' , 'wolf', 'tiger' , 'cat'}"
   ]
  },
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   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{'cat', 'dog', 'loin', 'rabbit', 'tiger', 'wolf'}"
      ]
     },
     "execution_count": 3,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "s2"
   ]
  },
  {
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   "execution_count": 4,
   "id": "64550d38-88ec-4412-9b40-e0b109e1ee49",
   "metadata": {},
   "outputs": [],
   "source": [
    "#s2[0]"
   ]
  },
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   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{1, 2, 3.1, 'Anshum'}"
```

```
      ]
     },
     "execution_count": 5,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "# multiple data types\n",
    "\n",
    "s3 = {1,2,3.1,'Anshum'}\n",
    "s3"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 6,
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   "metadata": {},
   "outputs": [],
   "source": [
    "# union \n",
    "# intersect\n",
    "# difference"
   ]
  },
  {
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   "execution_count": 7,
   "id": "fc4f81d5-74f0-48cd-801c-a4dc97fe38f9",
   "metadata": {},
   "outputs": [],
   "source": [
    "a = {1,2,3,4}\n",
    "b = {3,4,5,6}"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 8,
   "id": "4eb45992-91ae-465f-afec-871b5ab6652c",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "{1, 2, 3, 4, 5, 6}"
      ]
     },
     "execution_count": 8,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "#union\n",
    "a.union(b)"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 9,
   "id": "9d07ef7d-4199-45c4-b519-8b7a120fdb40",
```

```
    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "{3, 4}"
       ]
      },
      "execution_count": 9,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "#intersection\n",
     "\n",
     "a.intersection(b)"
    ]
   },
   {
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    "execution_count": 10,
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    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "{1, 2}"
       ]
      },
      "execution_count": 10,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# difference\n",
     "\n",
     "a.difference(b)\n",
     "a-b"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 11,
    "id": "f86f2bb9-9920-4737-ba29-eb7e94c775c1",
    "metadata": {},
    "outputs": [],
    "source": [
     "# update\n",
     "# intersection_update\n",
     "# difference_update"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 12,
    "id": "bbc14fe0-e62e-4b83-a2a1-0893316d4c36",
    "metadata": {},
    "outputs": [
     {
      "data": {
```

```
        "text/plain": [
         "{1, 2, 3, 4, 5, 6}"
        ]
       },
       "execution_count": 12,
       "metadata": {},
       "output_type": "execute_result"
      }
     ],
     "source": [
      "a = {1,2,3,4}\n",
      "b = {3,4,5,6}\n",
      "a.update(b)\n",
      "a"
     ]
    },
    {
     "cell_type": "code",
     "execution_count": 13,
     "id": "5de7cf41-f083-4af6-95f3-2e44b6fb00c5",
     "metadata": {},
     "outputs": [
      {
       "data": {
        "text/plain": [
         "{3, 4}"
        ]
       },
       "execution_count": 13,
       "metadata": {},
       "output_type": "execute_result"
      }
     ],
     "source": [
      "a = {1,2,3,4}\n",
      "b = {3,4,5,6}\n",
      "a.intersection_update(b)\n",
      "a"
     ]
    },
    {
     "cell_type": "code",
     "execution_count": 15,
     "id": "ad4dc790-fff7-4c11-b7b5-6fad1c41bb19",
     "metadata": {},
     "outputs": [
      {
       "data": {
        "text/plain": [
         "{1, 2, 3, 4}"
        ]
       },
       "execution_count": 15,
       "metadata": {},
       "output_type": "execute_result"
      }
     ],
     "source": [
      "a = {1,2,3,4}\n",
      "b = {3,4,5,6}\n",
      "a.difference_update\n",
      "a"
     ]
```

```
   },
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    "metadata": {},
    "outputs": [],
    "source": [
     "#add\n"
    ]
   },
   {
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    "execution_count": 17,
    "id": "3db284bb-05bb-4595-8ebc-f63d6559bf7e",
    "metadata": {},
    "outputs": [],
    "source": [
     "a = { 'manago','apple','bananas'}"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 18,
    "id": "7db37ede-1692-45ef-a22a-5d5afe6d2100",
    "metadata": {},
    "outputs": [
     {
      "name": "stdout",
      "output_type": "stream",
      "text": [
       "{'apple', 'bananas', 'manago'}\n"
      ]
     },
     {
      "data": {
       "text/plain": [
        "{'apple', 'bananas', 'kiwi', 'manago'}"
       ]
      },
      "execution_count": 18,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "print(a)\n",
     "a.add('kiwi')\n",
     "a"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 20,
    "id": "a0321ab8-32d7-4116-a077-32865a97a672",
    "metadata": {},
    "outputs": [
     {
      "name": "stdout",
      "output_type": "stream",
      "text": [
       "{'kiwi', 'apple', 'bananas', 'manago'}\n"
      ]
```

```
    },
    {
     "data": {
      "text/plain": [
       "{'bananas', 'kiwi', 'manago'}"
      ]
     },
     "execution_count": 20,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "#remove\n",
    "print(a)\n",
    "a.remove('apple')\n",
    "a"
   ]
  },
  {
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   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "set()"
      ]
     },
     "execution_count": 21,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "# clear\n",
    "a.clear()\n",
    "a"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 22,
   "id": "b52a6a42-fb90-4fd9-9aa4-088bc84c46db",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "set()"
      ]
     },
     "execution_count": 22,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "type({})\n",
    "\n",
    "#empty set\n",
```

```
    "\n",
    "s1 = set()\n",
    "s1"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 23,
   "id": "42f14b5a-bef2-49c3-9013-4ee149258607",
   "metadata": {},
   "outputs": [],
   "source": [
    "# issuperset\n",
    "# isssubset\n",
    "# isdisjoint"
   ]
  },
  {
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   "execution_count": 24,
   "id": "62352865-1713-4bf3-b7d9-ed28684726fe",
   "metadata": {},
   "outputs": [],
   "source": [
    "a = {1,2,3,4,5,6,7,8,9} # superset\n",
    "b = {3,2,1,4} #subset\n",
    "c = {5,6,10} # not a subject\n",
    "d = {10,20,30} # disjoint\n"
   ]
  },
  {
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   "outputs": [
    {
     "data": {
      "text/plain": [
       "False"
      ]
     },
     "execution_count": 25,
     "metadata": {},
     "output_type": "execute_result"
    }
   ],
   "source": [
    "a.issuperset(b)\n",
    "a.issuperset(c)"
   ]
  },
  {
   "cell_type": "code",
   "execution_count": 26,
   "id": "ea61b418-0bc5-4dd9-9ee3-9ee152539be7",
   "metadata": {},
   "outputs": [
    {
     "data": {
      "text/plain": [
       "False"
      ]
```

```
      },
      "execution_count": 26,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "b.issubset(a)\n",
     "b.issubset(c)"
    ]
   },
   {
    "cell_type": "code",
    "execution_count": 27,
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    "metadata": {},
    "outputs": [
     {
      "data": {
       "text/plain": [
        "False"
       ]
      },
      "execution_count": 27,
      "metadata": {},
      "output_type": "execute_result"
     }
    ],
    "source": [
     "# disjoint\n",
     "a.issuperset(b)\n",
     "d.isdisjoint(a)\n",
     "d.isdisjoint(c)"
    ]
   },
   {
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    "execution_count": null,
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    "metadata": {},
    "outputs": [],
    "source": []
   }
  ],
  "metadata": {
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    "language": "python",
    "name": "python3"
   },
   "language_info": {
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    "version": "3.13.9"
   }
  },
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